

Evidence-Based Study Strategy for the PhilNITS ITPEC FE Examination

Executive Summary

The pursuit of the Fundamental Information Technology Engineer (FE) certification—administered in the Philippines by the Philippine National IT Standards Foundation (PhilNITS) under the Information Technology Professional Examination Council (ITPEC) framework—represents a formidable challenge¹. For a fourth-year Bachelor of Science in Information Technology (BSIT) student currently engaged in an On-the-Job Training (OJT) internship, preparing for the October 25, 2026, examination within a highly constrained three-week window requires an aggressive, data-driven methodology. Time is the most critical constraint; therefore, traditional, exhaustive study methods must be discarded in favor of targeted cognitive triage.

This report provides a comprehensive, evidence-based study strategy engineered specifically for this operational reality. It acknowledges the candidate's distinct advantages—a strong foundational proficiency in Object-Oriented Programming (OOP) and Structured Query Language (SQL)—and immediately leverages these strengths to reclaim study hours. Concurrently, it addresses the candidate's primary vulnerability in abstract computer science theory by transforming conceptual learning into mechanical, formulaic execution. Furthermore, this analysis incorporates the sweeping structural revisions implemented in the April 2024 ITPEC syllabus, which modernized the examination into "Subject A" and "Subject B" and eliminated programming language selection in favor of a universal pseudocode².

By synthesizing pedagogical research, cognitive load theory, and empirical data from past ITPEC examinations, this document deconstructs the anatomy of the PhilNITS FE exam, delineates the precise boundaries between the Philippine ITPEC FE and the domestic Japanese IPA FE, and formulates a granular tactical approach to the algorithmic tracing required in Subject B⁴. The culmination of this strategic analysis is a rigorous, day-by-day 21-day schedule designed to maximize knowledge retention, optimize exam-day endurance, and secure the 600/1000 passing threshold required for both sections of the assessment⁶.

1. Exam Anatomy

To attack the FE examination efficiently, one must possess a forensic understanding of its structural mechanics, grading criteria, and recent syllabus modifications. It is a frequent error for candidates to conflate the ITPEC FE examination with the domestic Japanese FE examination administered by the Information-technology Promotion Agency (IPA). While both stem from the same foundational body of knowledge designed to validate the competencies of early-career software engineers, their delivery mechanisms, formats, and testing environments diverge significantly⁵.

Divergence Between ITPEC FE and Japan's Domestic IPA FE

The domestic Japanese IPA FE has evolved into a Computer-Based Testing (CBT) model,

permitting candidates in Japan to schedule and sit for the exam year-round in a digital environment⁷. It is administered exclusively in the Japanese language and requires an intermediate-to-advanced understanding of specialized Japanese technical vocabulary⁷. In stark contrast, the ITPEC FE, administered by PhilNITS in the Philippines, remains a traditional paper-and-pencil examination offered strictly on fixed dates twice a year—typically the last Sundays of April and October¹. The ITPEC FE is administered in English, aligning with the global market orientation of the ITPEC member nations (which include the Philippines, Thailand, Vietnam, Myanmar, Mongolia, and Bangladesh)¹. Consequently, candidates taking the PhilNITS exam must physically bubble an optical mark recognition (OMR) Scantron sheet using a pencil, which introduces mechanical time-management constraints and physical fatigue variables that CBT candidates do not encounter¹⁰.

The April 2024 Syllabus Overhaul

The candidate is preparing under the current ITPEC syllabus framework, which underwent a fundamental overhaul in April 2024². This revision abolished the legacy "Morning" and "Afternoon" nomenclature, replacing them with "Subject A" and "Subject B." This change structurally altered the pacing and volume of the assessment, as well as the nature of the technical evaluation.

Examination Parameter	Subject A (Formerly Morning)	Subject B (Formerly Afternoon)
Time Allocation	90 minutes (1.5 hours)	100 minutes (1.67 hours)
Total Questions	60 Multiple-Choice Questions	20 Scenario-Based Questions
Format & Selection	All 60 questions are mandatory	All 20 questions are mandatory
Pacing Requirement	1.5 minutes per question	5.0 minutes per question
Knowledge Domains	60% Technology, 40% Strategy & Management	Algorithms, Data Structures, Pseudocode, Cybersecurity

Prior to the 2024 revisions, the afternoon section permitted candidates to select a specific programming language (e.g., C, Java, Python, Assembly) and choose a subset of questions to answer. The current Subject B eliminates language selection entirely¹². All candidates must now read, interpret, and trace a universal ITPEC pseudo-programming language, leveling the playing

field but introducing a new syntactic framework that must be mastered⁴. Furthermore, Subject B now integrates cybersecurity scenarios requiring the analysis of access policies and threat vectors directly alongside algorithmic problem-solving².

Scoring Criteria, Pass Rates, and the Removal Policy

To achieve full certification, a candidate must score a minimum of 600 out of 1,000 points (60%) on **both** Subject A and Subject B⁵. The scoring algorithm does not employ a flat one-point-per-question metric; rather, it relies on weighted items based on question difficulty, calculated via Item Response Theory (IRT) or similar proprietary weighting mechanisms determined by the Japanese Question Making Committee⁹. Crucially, there is no penalty for incorrect answers¹⁵. This dictates an absolute exam-day rule: no question on the OMR sheet should ever be left blank.

The ITPEC FE is an objectively difficult examination. Historical data indicates that the pass rate in the Philippines is highly restrictive. For context, in the April 2024 ITPEC FE examination, 275 candidates tested in the Philippines, with only 42 passing, yielding a pass rate of 15.3%¹⁶. In April 2025, the pass rate was 21.8%, and in October 2025, it was 22.2%¹⁶. This establishes that the candidate is attempting to clear an assessment that fails roughly 80% of test-takers.

However, a critical strategic element of the PhilNITS framework is the "Removal Exam" policy. If a candidate passes one subject (e.g., Subject A) but fails the other (e.g., Subject B), the passing score is banked and remains valid for one year¹. The candidate is then permitted to take a "removal exam" for the failed section during the subsequent testing windows at a significantly reduced fee of PHP 1,200, compared to the standard PHP 2,500 full registration fee⁶. The implication for exam-day strategy is profound: if a candidate experiences catastrophic pacing issues or cognitive fatigue during Subject A in the morning, they must not surrender.

Proceeding to and passing Subject B secures that half of the credential, effectively halving the required study load for the next testing window⁶.

2. Triage for a 3-Week Window

With approximately 21 days until the October 25 examination, operating under severe time constraints due to an ongoing OJT internship, the candidate must optimize every available hour. Assuming 2.5 hours of study capacity on weekday evenings and 7 hours per day on weekends, the total available study budget is roughly 65 to 70 hours. Attempting to comprehensively consume the provided *IPA FE Exam Preparation Book Vols. 1 and 2*, which collectively span nearly 1,000 pages, is mathematically impossible and pedagogically flawed¹⁸. The Pareto Principle—identifying the 20% of effort that yields 80% of the results—must dictate all resource allocation.

Leveraging Existing Strengths (SQL and OOP)

The candidate's self-reported proficiency in Object-Oriented Programming (OOP) and SQL provides a massive tactical advantage that must be exploited. Subject A typically features multiple questions testing database normalization (1NF, 2NF, 3NF), relational algebra operations, transaction states (ACID properties), and complex SQL querying involving JOIN, GROUP BY, and HAVING clauses⁶. Furthermore, Subject B frequently utilizes OOP paradigms—such as

classes, instance instantiation, member variables, and method inheritance—within its pseudocode for data structures like Linked Lists and Binary Trees⁴.

Because the candidate is already strong in these domains, a strict triage directive must be implemented: **Zero primary study hours should be allocated to learning SQL syntax or basic OOP concepts.** The candidate should bypass the corresponding chapters in Volume 1 entirely. These topics will be sufficiently refreshed organically by encountering them in practice exams. Re-reading known material provides a false sense of productivity (fluency illusion) while stealing time from critical vulnerabilities.

Aggressive Prioritization of Vulnerabilities

Study time must be aggressively redirected toward the candidate's weaknesses: Computer Science theory and the mechanical execution of Subject B pseudocode. The triage hierarchy for the remaining 65 hours is structured as follows:

1. **Subject B Trace Execution (45% of time allocation):** Subject B constitutes 50% of the passing requirement but relies entirely on logical tracing rather than declarative memorization. Because the scenario-based questions carry heavy point weights, mastering the specific syntax of the ITPEC pseudo-language is the highest-yield activity¹².
2. **Subject A Calculations (25% of time allocation):** CS theory questions involving calculations (e.g., subnetting, memory access time, system reliability) are highly deterministic. Once the formula is understood, these questions become guaranteed points⁶.
3. **Strategy & Management Frameworks (20% of time allocation):** This domain constitutes a massive 40% of Subject A⁶. It tests the rote memorization of frameworks (Gantt, PERT, SWOT) and basic business accounting (Break-even points, ROI)¹⁹. Because it requires no deep technical intuition, it is easily memorized using spaced repetition and should not be neglected simply because it falls outside traditional computer science.
4. **Hardware and Low-Level CS Theory (10% of time allocation):** Deep theoretical dives into processor instruction pipelines, flip-flop circuits, and analog-to-digital quantization should be strictly limited. The candidate should only review high-frequency past-paper questions in this domain, accepting that some highly obscure hardware questions are statistically safer to guess.

The *IPA FE Exam Preparation Book Vols. 1 and 2* should be utilized exclusively as lookup encyclopedias. When a past paper question is missed, the candidate should consult the specific page in the textbook to clarify the concept, rather than reading the text sequentially.

3. Learning CS Theory as Someone Who Struggles With It

For a student who excels in applied programming but struggles with abstract computer science theory, the optimal pedagogical approach involves transforming theoretical concepts into mechanical, executable algorithms. The FE exam does not require candidates to write theoretical proofs or design circuit boards; it requires them to calculate specific outputs based on given inputs.

Because calculators are strictly prohibited in the exam room, all FE mathematical questions are engineered to cancel out cleanly or utilize easily reducible fractions¹¹. If a calculation results in an infinitely repeating decimal or requires complex long division, the candidate has likely applied the formula incorrectly. The strategy is to memorize the underlying algebra of high-frequency topics.

1. Memory Architecture and Average Access Time (AMAT)

Cache memory theory is a staple of Subject A²⁰. The fundamental concept is that processors check fast cache memory first; if the requested data is missing (a cache miss), the system must retrieve it from the slower main memory. The calculation for effective access time is highly deterministic:

$$AMAT = (H_c \times T_c) + (M_c \times T_m)$$

Where H_c is the Hit Ratio, T_c is Cache Access Time, M_c is the Miss Ratio ($1 - H_c$), and T_m is the penalty time (Main Memory Access Time). *Crucial nuance observed in ITPEC past papers:* If the question explicitly states that the main memory access time *includes* the time spent initially checking the cache, the formula does not require adding the cache time to the main memory time. Candidates must read the prompt carefully to determine if the main memory penalty is inclusive or exclusive of the cache check to avoid over-calculating the latency²⁰.

2. Reliability and Availability (MTBF and MTTR)

Systems engineering reliability is consistently tested²⁴.

- **Mean Time Between Failures (MTBF):** The average duration a system is operational.
- **Mean Time To Repair (MTTR):** The average duration a system is down for repairs.

Availability (the percentage of uptime) is calculated as:

$$\text{Availability} = \frac{\text{MTBF}}{\text{MTBF} + \text{MTTR}}$$

The exam frequently scales this by asking for the availability of systems arranged in series or parallel. For systems in series (where both must function), overall availability is the product of

individual availabilities ($A_1 \times A_2$). For redundant systems in parallel (where only one needs to function), overall availability is calculated by subtracting the probability of total simultaneous

failure from 1: $1 - ((1 - A_1) \times (1 - A_2))$.

3. Subnetting and IPv4 Architecture

IPv4 subnetting is a critical skill that bridges theory and network engineering²⁵. The candidate must mechanically map out Classless Inter-Domain Routing (CIDR) subnet masks.

- **Formula for New Subnets:** 2^n (where n is the number of borrowed network bits). For example, moving from a /24 to a /26 mask borrows 2 bits, creating $2^2 = 4$ subnets²⁶.
- **Formula for Usable Hosts:** $2^h - 2$ (where h is the remaining host bits). The subtraction of 2 is an absolute rule, accounting for the Network ID (all zeros) and the Broadcast Address (all ones), which cannot be assigned to individual devices²⁶.

4. Break-Even Point Analysis (Management Domain)

In the business strategy section, calculating the break-even point in sales volume is frequently tested¹⁹. The candidate must categorize expenses into Fixed Costs (costs that do not change regardless of volume, like rent) and Variable Costs (costs tied directly to production, like raw materials).

$$\text{Break-Even Revenue} = \frac{\text{Fixed Costs}}{1 - \left(\frac{\text{Variable Costs}}{\text{Sales Revenue}} \right)}$$

By treating business accounting as just another programmatic function with inputs and outputs, the candidate can secure points in a domain that heavily weights Subject A.

5. Radix Conversions and Logic Gates

The candidate must master the mechanical conversion of binary, octal, decimal, and hexadecimal numbers. For binary to hex, group the bits into blocks of four from right to left (e.g., $1011_2 \rightarrow B_{16}$)¹⁹. For logic gates, mechanical memorization of truth tables is required. For example, an AND gate outputs 1 strictly when all inputs are 1, while a NAND gate is the exact inverse⁶. De Morgan's laws are also frequently tested for boolean simplification:

$$\text{NOT (A AND B)} = (\text{NOT A}) \text{ OR } (\text{NOT B}) \quad ^6$$

By reducing these theoretical concepts to their underlying mathematical formulas and mechanical steps, the cognitive load is shifted from "understanding computer architecture" to "plugging values into an algebraic equation," playing directly into the strengths of a logic-oriented programmer.

4. The Pseudocode / Algorithm Section (Subject B)

Subject B represents the most significant hurdle of the modernized FE exam. Unlike traditional programming language tests, the ITPEC pseudo-language is designed to be language-agnostic. However, its specific notational rules present severe cognitive traps for students trained in modern C-family languages (Java, C#, JavaScript, C++). Success requires discarding assumptions and adhering strictly to the ITPEC syntax rulebook.

The 1-Based Indexing Trap

The most critical idiosyncrasy of the ITPEC pseudo-language is that **array indexing starts at 1, not 0**⁴. For a candidate well-versed in standard OOP, the instinct to start iterations at $i = 0$ is deeply ingrained in muscle memory. In ITPEC pseudocode, an array initialized as $\text{array} \leftarrow \{10,$

20, 30} places the integer 10 at array[1], not array[0].

Consequently, standard loop constructs are explicitly written to accommodate this: for (increase i from 1 to the number of elements in array by 1)⁴. Failing to mentally adjust to 1-based indexing will consistently lead to off-by-one errors during trace evaluations, resulting in incorrect outputs for sorting algorithms, matrix manipulations, and search functions.

The undefined State

The ITPEC framework introduces the concept of an undefined state, which functions similarly to null in Java or C#⁴. It is heavily utilized in questions involving dynamic data structures like linked lists and binary trees. For example, a linked list node representing the tail of the list will have its next property set to undefined. Attempting to access a property of an undefined element results in an execution error. Candidates must constantly monitor variable states to ensure they do not dereference an undefined pointer.

Short-Circuit Evaluation

The pseudo-language rigorously applies short-circuit evaluation for its logical operators (and, or)²⁹.

- **Logical AND (and):** If the first operand evaluates to false, the second operand is completely bypassed and ignored.
- **Logical OR (or):** If the first operand evaluates to true, the second operand is bypassed.

This mechanism is frequently tested in Subject B to ensure safe traversal of data structures. Consider the following common boundary check: if (node ≠ undefined and node.value == target) If node is currently undefined, the first condition evaluates to false. The execution engine immediately short-circuits, preventing the evaluation of node.value, which would otherwise trigger an illegal memory access or null-reference exception³¹. Recognizing this syntactic pattern alerts the candidate that the code is safely handling boundary conditions, which is crucial for answering questions about infinite loops or execution crashes.

Variable Assignment and Operators

The candidate must acclimatize to the specific notation. Assignment is denoted by a left-facing arrow ($\leftarrow$), not an equals sign. The equals sign (=) is reserved strictly for equality comparison¹². The remainder operation is denoted by mod, and logical negation is denoted by not¹².

The Trace Table Methodology

To conquer Subject B, reading code line-by-line using mental visualization is a recipe for catastrophic failure. Human working memory can only hold roughly 4 to 7 discrete items simultaneously; nested loops with multiple tracking variables (e.g., i, j, temp, pivot) will rapidly overflow this capacity, leading to tracking errors and incorrect answers.

The candidate must employ the **Trace Table** (or Desk Checking) methodology. On the scratch space of the exam booklet, the candidate should draw a grid where each column represents a specific variable in the algorithm.

1. As the code execution is traced line by line, the current value of the variable is written in the respective column.
2. When a variable changes state, the old value is crossed out with a single line, and the new value is written directly beneath it.

3. This creates a permanent, visual history of the algorithm's state at any given iteration²⁸. Trace tables neutralize the complexity of recursive functions and complex nested loops because they externalize the working memory onto paper, allowing the candidate to pause, analyze the logic, and resume without losing their place in the execution stack.

5. Past-Paper Strategy

In the final three weeks, relying on textbook reading is a sub-optimal strategy. The most robust findings in cognitive psychology regarding exam preparation highlight the "Testing Effect"—the phenomenon where the act of actively retrieving information from memory strengthens the neural pathway significantly more than passive reading. Therefore, past papers are not merely assessment tools; they are the primary learning materials¹⁸.

Focusing on Recent Vintages

Because the syllabus underwent a structural shift in April 2024, priority must be given to the most recent examinations (2024 to 2018). The ITPEC website freely publishes past questions and official answer keys³³. While the categorization changed to Subject A and Subject B, the core technological concepts remain identical. Older "Morning" exams map perfectly to the current Subject A. Older "Afternoon" algorithms—specifically the pseudocode, flowchart, and data structure questions—serve as excellent practice for Subject B, provided the candidate adapts them to the current universal pseudo-language format.

The Methodology of Distractor Analysis

It is insufficient to merely take a past paper and tally the final score. True learning occurs during the review phase through a process known as "Distractor Analysis." In a standard 4-option multiple-choice question, one answer is the key, and the remaining three are distractors. Exam authors do not generate distractors randomly. They are carefully engineered to reflect common misconceptions, arithmetic errors, or off-by-one algorithmic mistakes. When reviewing a practice exam, the candidate should analyze every question—even the ones answered correctly—and explicitly define *why* the other three options are incorrect²⁰.

- If option B in an MTBF question was derived by adding the variables instead of multiplying them, note that trap.
- If option C in a subnetting question represents the broadcast address rather than the total usable host range, document that trap.

By reverse-engineering the exam author's intent, the candidate builds an immunity to cognitive traps on the actual test day. An "Error Log" (maintained via a spreadsheet or notebook) should track every missed question, the root cause of the error (e.g., "forgot 1-based indexing," "confused SRAM with DRAM"), and the logical correction.

6. Memorization vs. Understanding

A successful triage strategy requires accurately distinguishing which concepts require deep conceptual understanding and which simply require brute-force declarative memorization. Treating both domains identically leads to wasted time and cognitive fatigue.

Deep Understanding (Procedural Knowledge)

Topics that require tracing, mathematical manipulation, and situational logic cannot be

memorized. These require deep procedural understanding. The examiner can change the variables, and the candidate must know how to apply the process. These include:

- Subnet mask calculations (CIDR notation, identifying network boundaries)²⁶.
- Data structure access policies (LIFO for Stacks, FIFO for Queues)⁶.
- Subject B algorithm tracing (sorting arrays, binary tree traversal, recursive factorial functions)⁴.
- Evaluating expressions in postfix (Reverse Polish) notation using stack operations⁶.

Rote Memorization (Declarative Knowledge)

Conversely, the Strategy and Management portions of Subject A, as well as specific hardware terminologies, represent declarative knowledge. There is no underlying logic to derive; the facts simply must be known. These include:

- Information security standards (ISO/IEC 27001)⁷.
- Project management tools (PERT charts, Critical Path Method, Gantt charts)⁶.
- Corporate strategy matrices (SWOT analysis, PPM matrix, Ansoff matrix, Value Chain analysis).
- System testing phases (Unit testing, Integration testing, System testing, Acceptance testing)³⁴.

Spaced Repetition Mechanics

For declarative knowledge, the candidate should utilize spaced repetition systems (SRS) like Anki. Rather than downloading massive, pre-made decks from the internet which contain irrelevant vocabulary³⁵, the candidate should create micro-flashcards exclusively for the concepts missed in past papers. If a past paper reveals a gap in understanding what an "Operational Amplifier" symbol looks like³⁶, that specific fact becomes a flashcard. Reviewing these flashcards during the daily commute or downtime of the OJT internship maximizes idle time, efficiently transferring declarative facts into long-term memory without consuming dedicated evening study blocks.

7. Exam-Day Tactics

Execution on exam day requires managing not just technical knowledge, but cognitive fatigue, severe time constraints, and strict environmental regulations. The FE exam is an endurance test lasting 190 minutes of active testing time, separated by a brief break¹.

Room Regulations and Logistics

The PhilNITS exam operates under strict manual testing conditions in designated university venues (e.g., in Cebu, Manila, or Davao)³⁷.

- **Writing Instruments:** Candidates must bring their own Mongol 2 (or standard HB/2B) pencils for shading the Scantron answer sheet, along with a high-quality, non-smudging eraser. A black ballpoint pen is required for filling out candidate information (name, examinee number)¹⁰.
- **Prohibited Items:** Calculators, smartwatches, smartphones, and any electronic devices are strictly prohibited¹¹. Attempting to use a calculator for the subnetting or MTBF formulas will result in immediate disqualification. Time is tracked by the proctors.

- **Documentation and Attire:** A valid ID and the official examination admission card are mandatory for entry¹¹. PhilNITS exam centers often enforce a dress code; candidates should wear a plain white polo or t-shirt with dark trousers, adhering to general professional testing standards³⁸.

Bubbling Strategy

Because the exam uses optical mark recognition (OMR) sheets, shading circles incorrectly, too lightly, or failing to erase mistakes completely can result in lost points despite the candidate knowing the correct answer¹⁰. The candidate should employ "batch-bubbling." Solve questions in the booklet, circling the answers on the paper, and then transfer the answers to the Scantron sheet in batches of 10. This minimizes the risk of offset errors (e.g., marking the answer for question 14 in the slot for question 15) and maintains the flow of cognitive problem-solving. Because there is no negative marking, any remaining blank circles must be guessed and filled in during the final two minutes of the session¹⁵.

Chronobiology and Fatigue Management

Because Subject A requires rapid, high-volume recall (1.5 minutes per question) and Subject B requires sustained, deep analytical tracing (5.0 minutes per question), the cognitive demands shift drastically. The candidate must aggressively utilize the break between the two subjects to rest their prefrontal cortex. This requires stepping away from the desk, hydrating, consuming a simple carbohydrate, and strictly avoiding any post-mortem discussions of Subject A with other candidates, which only induces anxiety and depletes remaining mental energy³⁸.

8. A Realistic 3-Week Schedule

Given the candidate's OJT schedule, this 21-day program assumes 2.5 hours of study per weekday evening and 7 hours per weekend day. It is divided into three distinct phases.

- **Phase 1 (Days 1-7): Deconstruction.** Focus entirely on mastering the mechanics of Subject B pseudocode and the mathematical formulas of Subject A. Use Volumes 1 & 2 only as reference guides.
- **Phase 2 (Days 8-14): Volume.** Massive exposure to past papers to build pattern recognition and stamina.
- **Phase 3 (Days 15-21): Synthesis.** Full timed simulations and rigorous Error Log review.

Date (2026)	Day	Focus Area	Specific Tactical Tasks
Oct 4	Sun	Subject B Foundation	Master ITPEC pseudocode rules (1-based arrays, undefined state, short-circuit rules). Practice trace table creation on 3 past

			algorithms.
Oct 5	Mon	Subject A Math	Memorize and drill Subnetting and IPv4 calculations. Practice solving without a calculator.
Oct 6	Tue	Subject A Math	Memorize and drill Cache AMAT, MTBF/MTTR, and System Availability formulas.
Oct 7	Wed	Subject B Drill	Trace 3 past Subject B algorithms (focus: Array manipulation, Stacks, and Postfix evaluation).
Oct 8	Thu	Subject A Theory	Review Boolean Algebra (De Morgan's laws), Binary/Hex conversions, and Logic Gates.
Oct 9	Fri	Subject A Strategy	Memorize Strategy/Management frameworks (PERT, SWOT, ROI, Break-even). Add missed terms to Anki.
Oct 10	Sat	Simulation 1	Take 2024 April Subject A (Full 90 mins). Spend 3 hours conducting

			Distractor Analysis and updating the Error Log.
Oct 11	Sun	Simulation 2	Take 2024 April Subject B (Full 100 mins). Use trace tables for every question. Spend 3 hours analyzing logic failures.
Oct 12	Mon	Weakness Targeting	Review Error Log from Simulation 1. Drill 50 flashcards. Consult Vol 1/2 for specific missed concepts.
Oct 13	Tue	Subject B Drill	Trace 3 past Subject B algorithms (focus: Linked Lists & Binary Trees).
Oct 14	Wed	Subject A Theory	Review Hardware architecture, Memory types (SRAM vs. DRAM), and basic CPU pipelining.
Oct 15	Thu	Subject A Theory	Review basic Security concepts (ISO 27001, PKI, Asymmetric Encryption types).
Oct 16	Fri	Mixed Drill	Light review of SQL (JOINS) and OOP

			concepts just to refresh syntax.
Oct 17	Sat	Simulation 3	Take 2023 October Subject A (Full 90 mins). Conduct Distractor Analysis and update Error Log.
Oct 18	Sun	Simulation 4	Take 2023 October Subject B (Full 100 mins). Conduct Distractor Analysis and update Error Log.
Oct 19	Mon	Weakness Targeting	Isolate the top 3 recurring failure types in the Error Log and drill them exclusively.
Oct 20	Tue	Subject B Speed	Trace 3 complex Subject B algorithms against a strict 15-minute clock per scenario.
Oct 21	Wed	Subject A Speed	Take 30 random Subject A past questions against a strict 45-minute clock.
Oct 22	Thu	Error Log Mastery	Do not learn new material. Reread the entire Error Log. Clear all Anki flashcard queues.

Oct 23	Fri	Logistics Prep	Pack Mongol 2 pencils, black pen, valid ID, admission ticket. Do light 1-hour review.
Oct 24	Sat	Rest	Zero studying. Pre-hydrate. Prioritize 8+ hours of sleep to ensure cognitive readiness.
Oct 25	Sun	EXAM DAY	Execute bubbling strategy. Triage hard questions. Use the break to reset.

List of Sources

- [cite: 2] PhilNITS Foundation, "FE & IP Exam Changes April 2024."
- [cite: 6] OpenExamPrep, "PhilNITS FE Exam Structure, Syllabus, and Practice Questions."
- [cite: 18] Kradasa, "How to Prepare for the ITPEC FE Exam," Dev.to.
- [cite: 36] Scribd, "2025S FE-A Questions."
- [cite: 17] PhilNITS Foundation, "Fundamental Information Technology Engineers (FE) Certification Exams."
- [cite: 1] Duran, Irving, "Fundamental Information Technology Engineer (FE) Exam: A Pathway to Engineering in Japan."
- [cite: 3] ITPEC.org, "Outline of ITPEC Common Examination from April 2024."
- [cite: 4] ITPEC.org, "Sample Questions for FE Subject B (Pseudo programming language notations)."
- [cite: 12] Scribd, "2025S FE-B Questions."
- [cite: 29] Scribd, "2024A FE-B Questions."
- [cite: 13] ITPEC Studio, "ITPEC FE Exam Topic Breakdown."
- [cite: 2] PhilNITS Foundation, "PhilNITS FE exam structure changes April 2024."
- [cite: 6] OpenExamPrep, "ITPEC FE exam passing rate Philippines."
- [cite: 16] ITPEC.org, "Examination Statistics (2019-2026)."
- [cite: 5] Wikipedia, "Information Technology Professional Examination Council."
- [cite: 6] OpenExamPrep, "PhilNITS FE removal exam rules."
- [cite: 6] OpenExamPrep, "PhilNITS FE calculation questions."
- [cite: 25] IPCisco, "Subnetting Examples."
- [cite: 26] The Power Education, "Subnetting: What is it."

- [cite: 24] Scribd, "April 2016 PHILNITS (MTBF MTTR)."
- [cite: 10] Scribd, "Exam Answer Sheet Instructions."
- [cite: 34] PhilNITS Davao, "PhilNITS Flyer v0-2."
- [cite: 11] 3D Universal, "Test-Taking Rules and Prohibited Items."
- [cite: 20] ITPECPpractice.com, "Cache Hit Rate and Average Access Time."
- [cite: 23] ITPECPpractice.com, "Average memory access time cache hit ratio formula."
- [cite: 19] IPA, "FE Exam Preparation Book VOL1."
- [cite: 27] Scribd, "Answers of Exercises of Vol 1 and Vol 2."
- [cite: 21] IPA, "Sample usage simple calculation of a break-even point."
- [cite: 19] IPA, "FE Exam Preparation Book VOL1 (Strategy)."
- [cite: 33] ITPEC.org, "Past Examination Questions and Answers."
- [cite: 9] ITPEC.org, "Features of the Examination."
- [cite: 35] Reddit (r/LearnJapanese), "How should I study for IT Passport."
- [cite: 28] Scribd, "2025S to 2021A FE-PM Q-A (Trace Tables)."
- [cite: 1] Duran, Irving, "ITPEC FE vs Japan IPA FE."
- [cite: 5] Wikipedia, "ITPEC Certifications."
- [cite: 7] TokyoDev, "The Fundamental Information Technology Engineer Examination: My Experience."
- [cite: 9] ITPEC.org, "Question Making Committee."
- [cite: 28] Scribd, "Pseudo programming language undefined array 1-based indexing rules."
- [cite: 4] ITPEC.org, "FE Subject B Pseudocode Notations."
- [cite: 31] CodersLegacy, "Short Circuit Evaluation in Programming."
- [cite: 32] Dev.to, "Short circuit evaluation."
- [cite: 30] Medium, "You must know the short circuit evaluation in Java."
- [cite: 6] OpenExamPrep, "PhilNITS FE pass rate breakdown."
- [cite: 15] PPI2Pass, "FE Exam Pass Rates."
- [cite: 14] Reddit (r/FE_Exam), "Do you need to pass every subject to pass the FE."
- [cite: 38] Scribd, "P2 Examination Reminders."
- [cite: 22] Scribd, "List of Allowed and Disallowed Items."
- [cite: 39] Scribd, "Handbook-002."
- [cite: 37] Facebook (PhilNITS), "Exam Venue Locations."
- [cite: 4] ITPEC.org, "FE Subject B pseudocode binary tree traversal."
- [cite: 8] Wikipedia, "PhilNITS."

Works cited

1. Fundamental Information Technology Engineer (FE) Exam, <https://duranirving.com/articles/fundamental-it-engineer-exam-japan>
2. FE & IP Exam Changes April 2024 - philnits.org, <https://philnits.org/common-body-79/>
3. ITPEC.org | IT Professionals Examination Council, <https://itpec.org/>
4. Sample Questions - ITPEC.org, <https://itpec.org/about/FE-B-SampleQuestions.pdf>

5. Information Technology Professional Examination Council - Wikipedia, https://en.wikipedia.org/wiki/Information_Technology_Professional_Examination_Council
6. Free Practice Questions for PhilNITS FE | OpenExamPrep, <https://open-exam-prep.com/practice/ph-philnits-fundamental-it-engineer>
7. The Fundamental Information Technology Engineer Examination, <https://www.tokyodev.com/articles/no-degree-take-the-it-exam-that-can-get-you-a-japanese-engineer-visa>
8. PhilNITS - Wikipedia, <https://en.wikipedia.org/wiki/PhilNITS>
9. About ITPEC Common Examination, <https://itpec.org/about/itpec-common-exam.html>
10. Exam Answer Sheet Instructions | PDF | Pencil | Tools - Scribd, <https://fr.scribd.com/document/636508292/ANSWER-SHEETS>
11. Test-Taking Rules and Prohibited Items: Civil Service Exam Guide, <https://3d-universal.com/en/blogs/test-taking-rules-and-prohibited-items.html>
12. IT Engineer Exam Instructions & Pseudocode | PDF - Scribd, <https://www.scribd.com/document/886508994/2025S-FE-B-Questions>
13. FE Practice Test - ITPEC Studio, <https://itpecstudio.com/FE>
14. Do you need to pass every subject to pass the FE? : r/FE_Exam, https://www.reddit.com/r/FE_Exam/comments/1amwi8q/do_you_need_to_pass_every_subject_to_pass_the_fe/
15. 2025-2026 FE Exam Pass Rates - PPI2Pass, <https://ppi2pass.com/resources/fe-exam/pass-rates>
16. Examination Statistics | ITPEC.org, <https://itpec.org/statsandresults/examination-statistics.html>
17. philnits.org - Philippines, <https://philnits.org/>
18. How to Prepare for the ITPEC FE Exam - DEV Community, <https://dev.to/kradasa4/how-to-prepare-for-the-itpec-fe-exam-4jbi>
19. FE Exam Preparation Book VOL1 LimitedDisclosureVer | PDF - Scribd, <https://www.scribd.com/doc/7341578/FE-Exam-Preparation-Book-VOL1-LimitedDisclosureVer>
20. ITPEC FE Morning April 2018 Question 10 — Answer & Explanation, <https://itpecpractice.com/fe/2018-march/morning/question-10>
21. IT Passport Examination (Level 1) - IPA, <https://www.ipa.go.jp/en/it-examinations/nph2g600000007uh-att/000009636.pdf>
22. Exam Room Item Guidelines | PDF - Scribd, <https://www.scribd.com/document/857625453/List-of-Allowed-and-Disallowed-Items>
23. ITPEC FE Morning October 2020 Question 10, <https://itpecpractice.com/fe/2020-october/morning/question-10>
24. April 2016 PHILNITS | PDF | Information Security | Business Process, <https://www.scribd.com/document/329458988/April-2016-PHILNITS>
25. Subnetting Examples (With Questions & Answers) - IPCisco, <https://ipcisco.com/lesson/subnetting-examples/>

26. Subnetting: What It Is, How to Calculate It & Why It Matters,
<https://thepower.education/en/blog/subnetting-what-is-it>
27. Commentaries and Answers Exercises: New FE Textbook - Scribd,
<https://www.scribd.com/document/465171070/Answers-of-Exercises-of-Vol-1-and-Vol-2-pdf>
28. ITPEC Fundamental IT Engineer Exam 2025 | PDF - Scribd,
<https://www.scribd.com/document/927786283/2025S-to-2021A-FE-PM-Q-A>
29. IT Engineer Exam Instructions | PDF | Key (Cryptography) - Scribd,
<https://www.scribd.com/document/878342721/2024A-FE-B-Questions>
30. You must know The Short Circuit Evaluation in Java - Medium,
<https://pradeesh-kumar.medium.com/you-must-know-the-short-circuit-evaluation-in-java-279db26ef3a5>
31. Short Circuit Evaluation in Programming - CodersLegacy,
<https://coderslegacy.com/short-circuit-evaluation-in-programming/>
32. Short circuit evaluation - DEV Community,
<https://dev.to/coderjay06/short-circuit-evaluation-3jb1>
33. FE Past Examination Questions and Answers - ITPEC.org,
<https://itpec.org/pastexamqa/fe.html>
34. Philippine National IT Standards (PhilNITS) Certification Exam,
http://philnitsdavao.weebly.com/uploads/2/5/0/3/2503911/philnits_flyer_v0-2_.pdf
35. How should I study for IT Passport?(or any other proficiency tests held,
https://www.reddit.com/r/LearnJapanese/comments/17w00nd/how_should_i_study_for_it_passportor_any_other/
36. Fundamental IT Engineer Exam 2025 | PDF | Domain Name System,
<https://www.scribd.com/document/886508842/2025S-FE-A-Questions>
37. WHAT : PHILNITS/ITPEC NATIONWIDE EXAM FOR IP (Level-1) , FE,
<https://www.facebook.com/100057143242771/posts/what-philnitsitpec-nationwide-exam-for-ip-level-1-fe-level-2when-april-26-2026-s/1417019600212801/>
38. Board Exam Preparation Guidelines | PDF | Clothing - Scribd,
<https://www.scribd.com/document/906436279/P2-Examination-Reminders>
39. PhilNITS IT Exam Handbook 2023 | PDF | Business - Scribd,
<https://www.scribd.com/doc/96507699/Handbook-002>